

RAD

The Running Analysis Device

A foot-mounted wearable that turns any runner into a lab subject — for under \$60.

HUDSON HOWLAND & BRANTLEY FIELD

PLTW ENGINEERING CAPSTONE



Running is breaking the people who love it.

79%

of recreational runners get injured every year. (*NIH*)

Unknown Cause

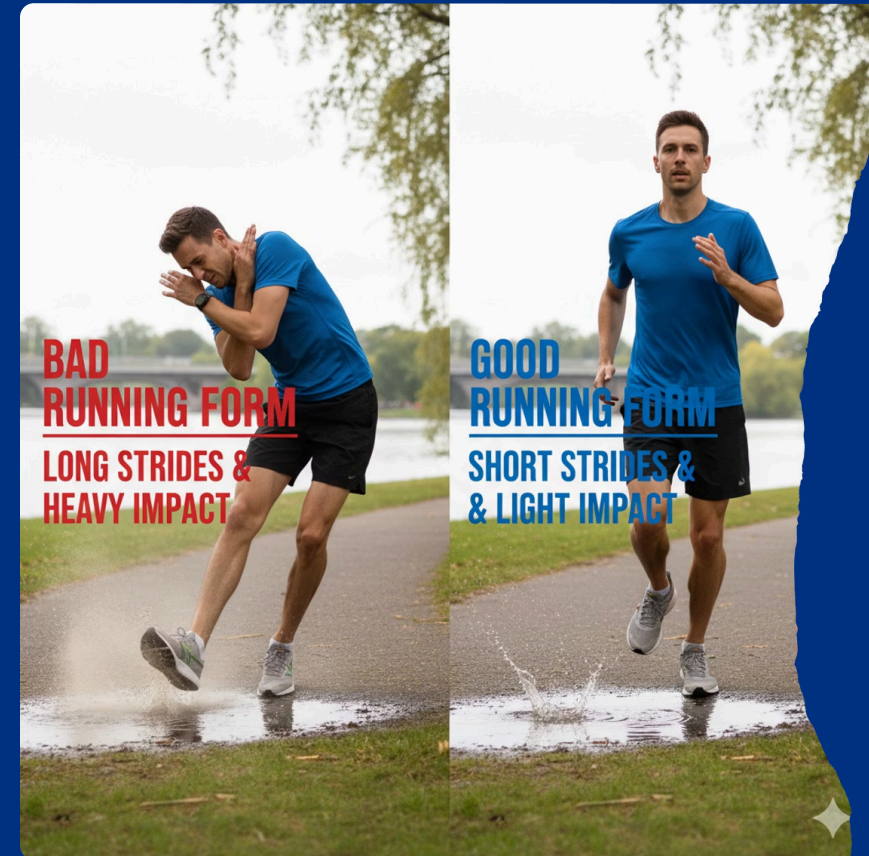
Most don't know why they get injured

No Diagnosis

Most never get properly diagnosed

They Quit

Most just stop running altogether



It's not bad luck. It's bad form.

60%

Overuse Injuries

50–70% of running injuries stem
from poor biomechanics

30×

Risk Multiplier

Jumping mileage without fixing
form raises injury risk 20–40×

10%

Cadence Fix

A 5–10% cadence increase
measurably cuts knee and hip
loading

Sources: JOSPT / NIH

The fix exists. It's locked behind a paywall.

\$350—\$500

per session for professional lab gait analysis.

→ Full Setup Required

Treadmill, motion capture, and a sports scientist — every time

→ One-Time Snapshot

Not ongoing feedback — a single session tells you nothing about your progress

→ Out of Reach

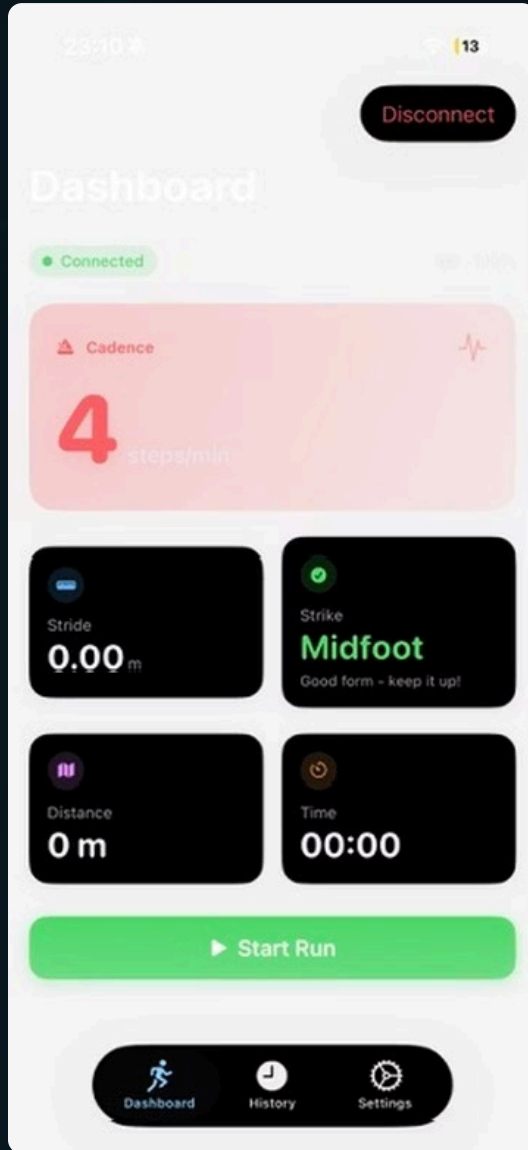
Completely inaccessible for high school athletes and recreational runners



The competition isn't competing.

Product	Price	Why it fails
RunScribe	\$600	Out of business
Stryd 5.0	\$200	Uses "power" — a metric most runners can't interpret or act on
Lab Gait Analysis	\$350– \$500/session	Not portable, not accessible, not ongoing
RAD	<\$60	Real metrics. On the shoe. Under sixty bucks.

Every competitor either costs too much, went out of business, or delivers data runners can't use. RAD is the only device that hits all three marks: affordable, wearable, and actionable.



INTRODUCING

Meet RAD.

Lab-grade gait feedback. On your shoe. Under \$60.

Wireless Streaming

Foot-mounted wearable streams gait data over Bluetooth in real time

Live iPhone App

Shows cadence, stride, strike type, and distance — live on your phone

Capstone Built

Designed and engineered in a high school engineering capstone

Built for the runners no one's measuring.



Recreational Runners

Chasing PRs without breaking down — they need data, not guesswork



HS Cross-Country Athletes

No access to sports science — 60% surveyed didn't know their cadence. 75% feared injury.



Coaches

Want objective data instead of eyeballing form from the sideline

Five metrics. Every step. In real time.



Cadence

Steps per minute, sampled continuously



Stride Length

Meters per stride via inverted pendulum model



Foot Strike

Heel / mid / forefoot classification per step



Distance

Cumulative distance tracked in real time



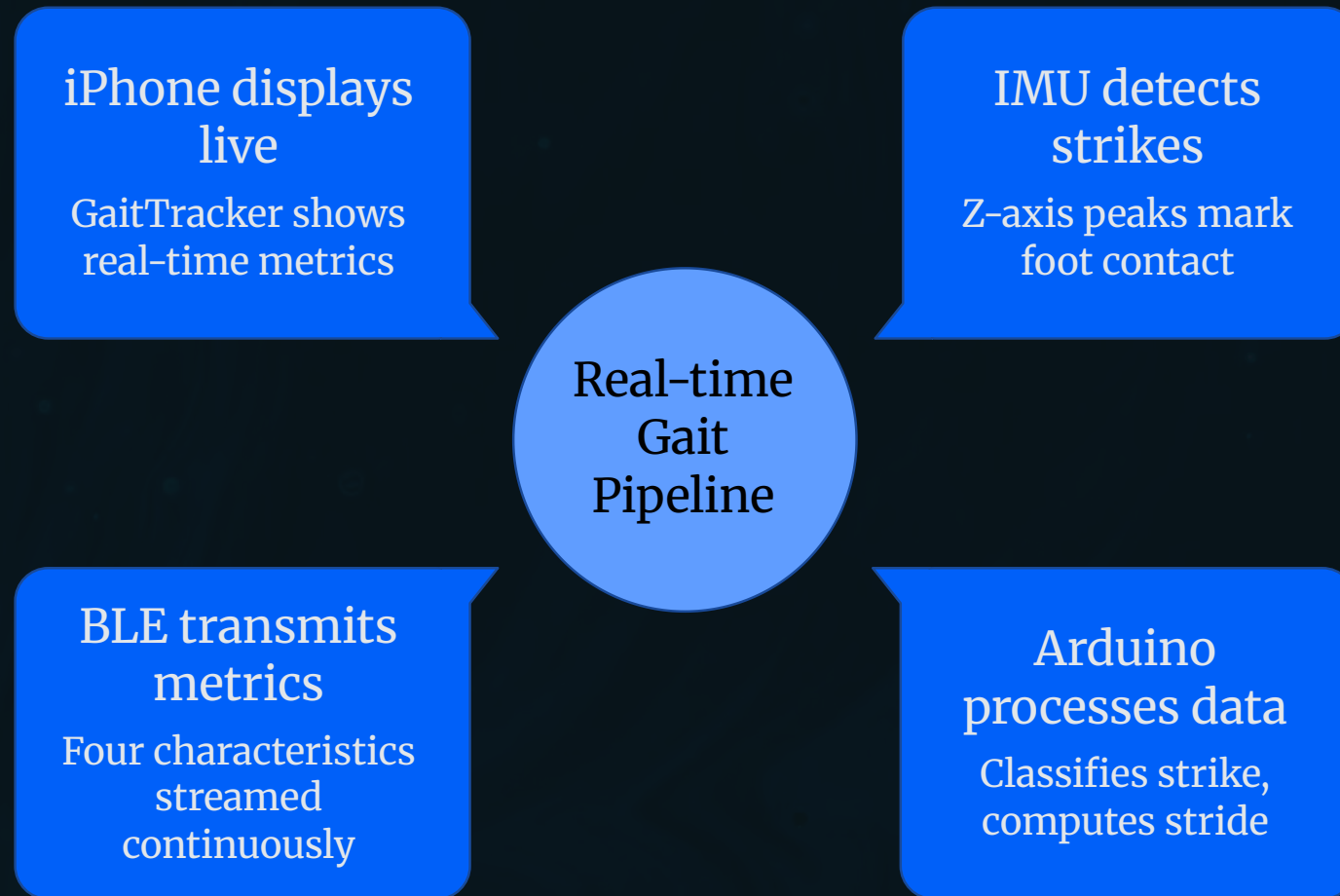
Battery

Live battery status streamed over BLE

 Sampled at **100 Hz**, transmitted continuously over Bluetooth Low Energy

How it works.

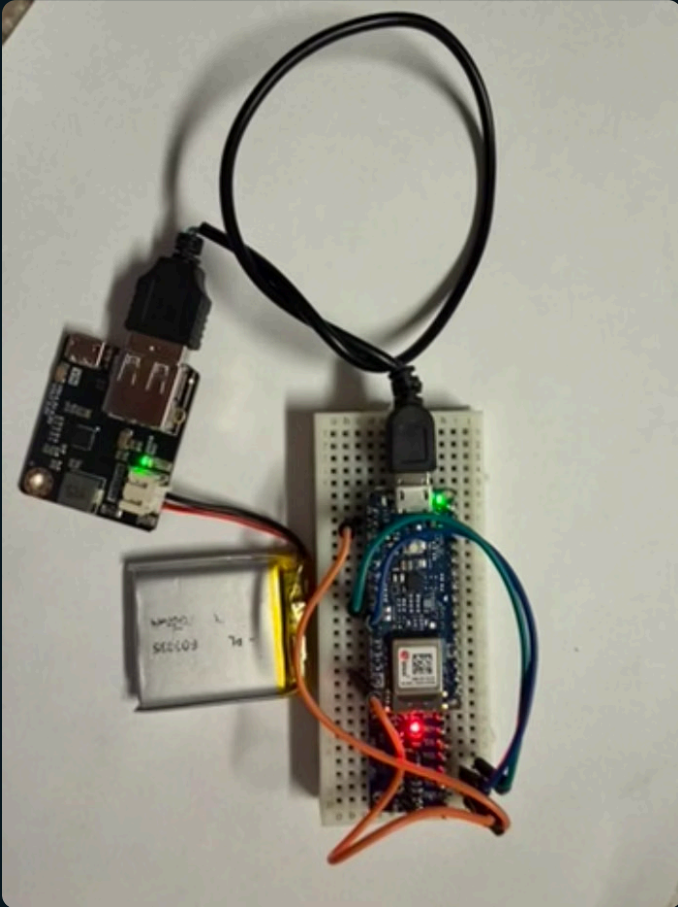
Sensor → MCU → BLE → iPhone



The IMU detects foot strikes from Z-axis acceleration peaks. The Arduino classifies strike type by pitch angle and computes stride via the inverted pendulum model. Four BLE characteristics stream continuously to the GaitTracker app on iPhone.

HARDWARE

The hardware. ~\$55 BOM



Micro- controller

Arduino Nano 33
BLE Rev2

Motion Sensor

MPU-6050 IMU

Power

MP2636 module ·
3.7V 700mAh
LiPo

Enclosure

3D-printed PLA snap-fit · 50 × 40 × 20 mm — laces onto any shoe

The firmware.



100 Hz Sampling

200ms
debounce
window
eliminates false
strike detections
from noise

Strike Classification

Heel $< -15^\circ$ ·
Forefoot $> +5^\circ$ ·
Midfoot
between —
classified per
step by pitch
angle

Stride Equation

Inverted
pendulum
model:
$$\text{stride} = 4 \times$$
$$\text{legLength} \times$$
$$\sin(\text{angle} / 2)$$

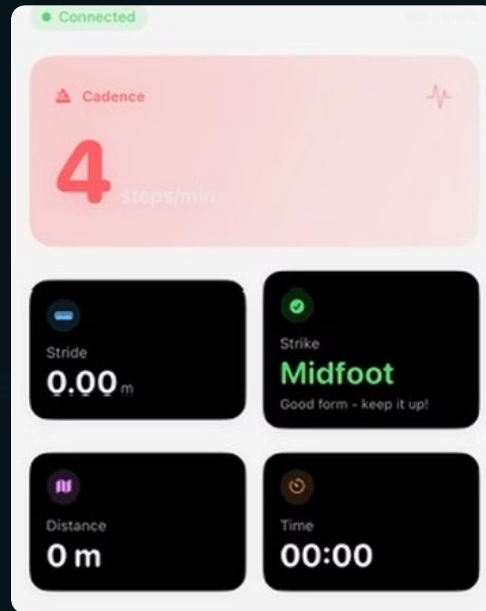
The app.

GaitTracker — Swift / SwiftUI



Connect

Pulsing BLE rings guide you through pairing the device to your phone in seconds



Live Dashboard

Cadence hero card shifts color as you cross the optimal zone — real-time feedback every step



History & Share

Bar charts for cadence and distance, donut for strike type, plus auto-generated PNG share card

Real-World Test Results

1.40 km

Distance Recorded

Device measurement (actual: 0.84 km)

1,157

Steps Counted

Total steps during half-mile walk

14:10

Duration

Time elapsed during test

74 spm

Cadence

Average steps per minute

Forefoot

Strike Type

Predominant foot strike detected

Distance Accuracy

The device overestimated distance by approximately 67%. This is a known limitation of the inverted pendulum stride model and will be corrected by tuning the stride correction factor against measured track data.

Strike Classification

The device recorded nearly all steps as forefoot strikes due to the device being mounted at a forward angle on the shoe. A firmware calibration update was written to automatically detect and subtract the mounting angle offset at startup, correcting this issue.

Core functionality confirmed working. Both issues identified have known fixes and are planned for the next iteration.

Stakeholder Feedback

Jackson
Peer / Athlete

"This could be useful for running. It would help you know when you're doing good or bad, which would help change your running form for the better. Maybe add padding to make it not a plastic box on your ankle and less corners for a more slick design."

Trey Howland
Software Engineer

"Impressive first prototype. I would like to see realtime data as a widget and historical data as a graph. I would want to know that the data is secure and use it for a while to make sure it's reliable before recommending it to someone."

Diana
Ex Competitive
Runner / Junior
Olympic Athlete

"At under \$60 I'd suggest it to any high school runner who can't afford lab gait analysis. The idea is right, the execution just needs another iteration."

All three stakeholders validated the core concept and identified the same priorities for improvement: smaller form factor, comfort, and accuracy validation.

Built to a spec sheet. Hit the spec sheet.

Priority	Target	Status
Cost	BOM under \$60	✓ MET
Size	50 × 40 × 20 mm enclosure	✓ MET
Data Rate	≥100 Hz sampling	✓ MET
Cadence Accuracy	Within 3 spm	● Pending field test
Battery Life	≥3 hours continuous	● Pending test
Durability	IPX3 moisture resistance	● Pending test

✔ Three of six primary specs confirmed met. Field testing underway for remaining criteria.

Where we are right now.



Firmware — Done

All 4 BLE characteristics transmitting reliably at 100 Hz



iOS App — Done

Running on a physical iPhone — all features live including history and share card



Enclosure — Done

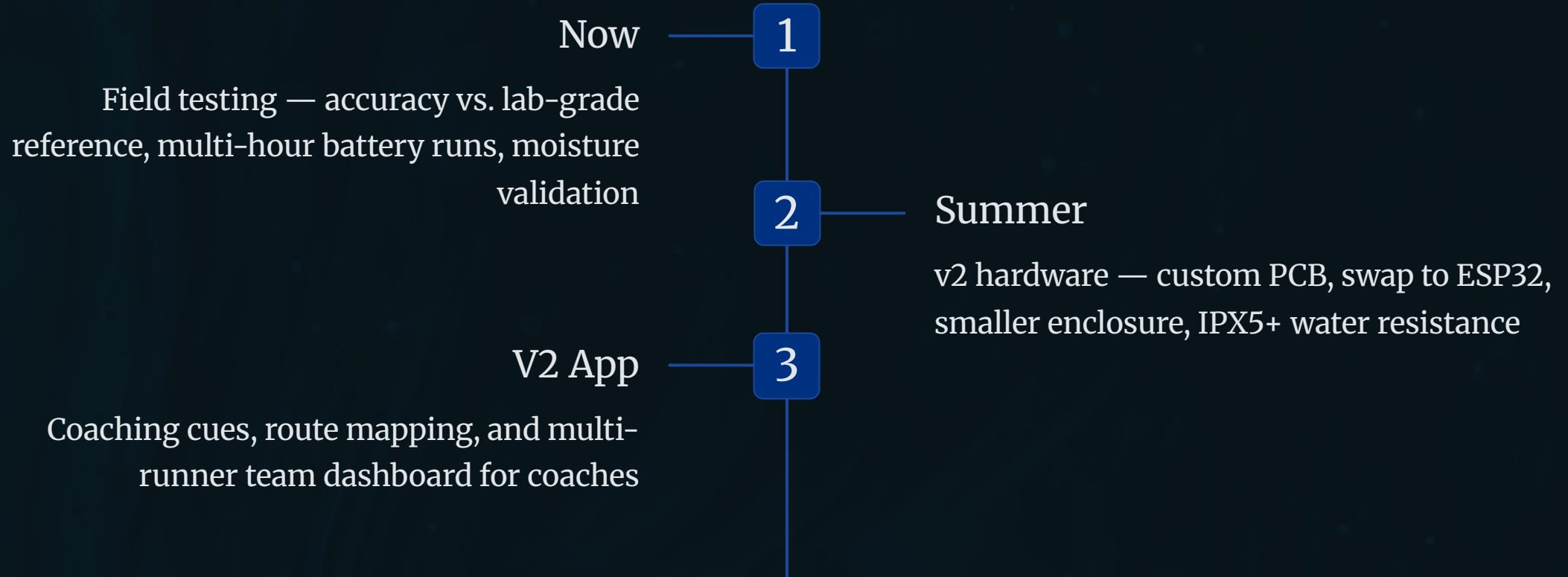
Blue PLA snap-fit enclosure mounted to a shoe and field-ready



Pending: cadence accuracy vs. metronome reference, battery life validation, IPX3 moisture resistance test.



What's next.



The roadmap is clear: validate the current build, miniaturize the hardware, and expand the software into a full coaching platform accessible to every team.

Lab-grade gait analysis made simple, easy, and cheap.

\$60

RAD makes the data that prevents injuries available to every runner — not just the ones who can afford a lab.

Hudson Howland & Brantley Field · PLTW 2026

Sources: NIH · Journal of Orthopaedic & Sports Physical Therapy · Internal HS XC survey

RAD — THE RUNNING ANALYSIS DEVICE